WORKED SOLUTION

STEP I 2015, Question 11

Mechanics • Friction and Moments

STEP I 2015, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s1-q11

The key idea

Every normal reaction on a circular cylinder acts along a radius, so it passes through the centre. Taking moments about each centre therefore kills the weight and all the normal reactions, leaving only the frictional forces, which all act at the same distance from the centre. That gives part (i) almost for free. For part (ii), resolve along the plane for each cylinder, then apply the friction law at the two contacts that the inequalities concern.

Solution

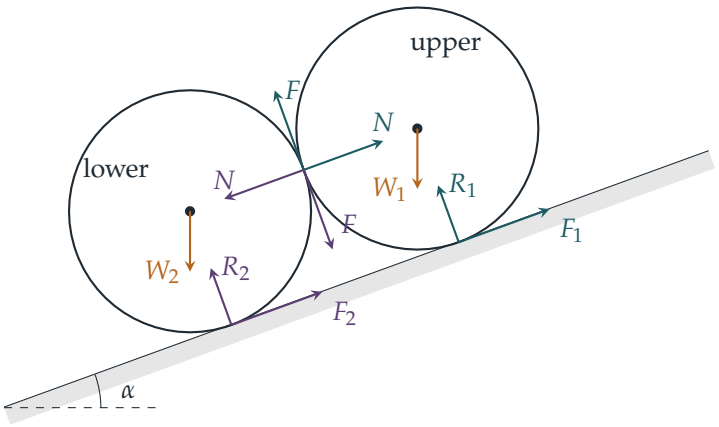
We model the cylinders as uniform rigid cylinders of radius r, both touching the plane, with the upper one higher up the slope. Since the radii are equal, the line of centres is parallel to the plane, so the point where the cylinders touch is at height r above the plane, level with both centres. The normal force between the cylinders acts along the line of centres (parallel to the plane), and the friction between them is perpendicular to the plane.

Take the x-direction up the slope and the y-direction perpendicular to the plane, away from it. Let

- R_1, R_2 be the normal reactions of the plane on the upper and lower cylinders;
- F_1, F_2 be the frictional forces of the plane on the upper and lower cylinders, measured *up* the slope (we let these be signed, so a negative value would mean down the slope);
- N be the normal force of the lower cylinder on the upper one, which acts up the slope;
- F be the friction of the lower cylinder on the upper one, measured *away from* the plane (again signed).

By Newton's third law the upper cylinder exerts N down the slope and F towards the

plane on the lower cylinder.



Forces on the upper cylinder are in teal and those on the lower cylinder in plum.

Part (i)

Take moments about the centre of the upper cylinder. Its weight acts at the centre, and R_1 and N both act along radii, so none of them has a moment. Only the two frictional forces remain, each acting along a tangent at distance r from the centre:

- F_1 acts up the slope at the lowest point of the cylinder, giving moment rF_1 anticlockwise (in the picture);
- F acts away from the plane at the point of contact, which is on the down-slope side of the centre, giving moment rF clockwise.

For equilibrium these balance, so $rF_1 = rF$, that is, $F_1 = F$.

Ask yourself Why can we ignore the normal reactions when we take moments about the centre?

A normal reaction on a circle is perpendicular to the tangent at the point of contact, so it lies along the radius and its line of action passes through the centre. Its moment about the centre is zero. This is exactly why moments about the centres are the right choice here.

For the lower cylinder, in the same way, only F_2 (up the slope at its lowest point, moment rF_2 anticlockwise) and the friction F towards the plane at the point of contact, which is on the up-slope side of its centre (moment rF clockwise) have moments. So $F_2 = F$.

Hence $F = F_1 = F_2$ as signed quantities, and in particular their magnitudes are equal, as required.

Part (ii)

Resolve parallel to the plane for each cylinder:

$$\begin{aligned} \text{upper:} \quad & F_1 + N = W_1 \sin\alpha, \\ \text{lower:} \quad & F_2 = N + W_2 \sin\alpha. \end{aligned}$$

Adding, and using $F_1 = F_2 = F$,

$$2F = (W_1 + W_2)\sin\alpha, \qquad \text{so} \qquad F = \tfrac{1}{2}(W_1 + W_2)\sin\alpha > 0,$$

which also tells us that every friction force acts in the direction we drew it. Substituting back,

$$N = W_1 \sin\alpha - F = \tfrac{1}{2}(W_1 - W_2)\sin\alpha,$$

which is positive because $W_1 > W_2$ and $\alpha > 0$, so the cylinders really do press on each other. Resolving perpendicular to the plane for the upper cylinder,

$$R_1 + F = W_1 \cos\alpha, \qquad \text{so} \qquad R_1 = W_1 \cos\alpha - F.$$

Ask yourself Which friction inequalities do we actually need?

In equilibrium, friction at each contact can be at most the coefficient of friction times the normal reaction there. The first result comes from the contact between the cylinders, and the second from the contact between the upper cylinder and the plane. (The lower contact gives a third condition, involving μ_2, which isn't asked for.)

Between the cylinders. We need $F \le \mu N$:

$$\tfrac{1}{2}(W_1 + W_2)\sin\alpha \le \mu \cdot \tfrac{1}{2}(W_1 - W_2)\sin\alpha.$$

Dividing by $\tfrac{1}{2}(W_1 - W_2)\sin\alpha$, which is positive, gives

$$\mu \ge \frac{W_1 + W_2}{W_1 - W_2},$$

as required.

Upper cylinder and plane. We need $F \le \mu_1 R_1 = \mu_1(W_1 \cos\alpha - F)$, that is,

$$(1+\mu_1)F \le \mu_1 W_1 \cos\alpha \quad \iff \quad \tfrac{1}{2}(1+\mu_1)(W_1+W_2)\sin\alpha \le \mu_1 W_1 \cos\alpha.$$

The angle of an inclined plane satisfies $0 < \alpha < \tfrac{1}{2}\pi$, so $\cos\alpha > 0$, and $(1+\mu_1)(W_1 + W_2) > 0$. Dividing by both,

$$\tan\alpha \le \frac{2\mu_1 W_1}{(1+\mu_1)(W_1+W_2)},$$

as required. (Note that this also forces $R_1 > 0$, since $\mu_1 R_1 \ge F > 0$, so the upper cylinder stays on the plane.)

Checks

- With $W_1 = 3$, $W_2 = 1$ we need $\mu \geq 2$, and with $\mu_1 = 1$ as well we need $\tan\alpha \leq \frac{3}{4}$. Substituting such values into the six equilibrium equations (two resolutions and one moment equation per cylinder) and solving them directly gives the same F, N and R_1 as above.
- As $W_1 \to W_2$ the bound on μ goes to infinity. That makes sense: then $N \to 0$, so the cylinders barely press on each other, yet the friction between them must still be $\frac{1}{2}(W_1 + W_2)\sin\alpha$.
- As $\mu_1 \to \infty$ the bound becomes $\tan\alpha \leq 2W_1/(W_1 + W_2)$, which is exactly the condition $R_1 \geq 0$: even a perfectly rough plane can't help once the upper cylinder starts to lift off.

Common slips

- Taking the friction between the cylinders to act along the slope. It acts along the common tangent at the contact, which is perpendicular to the plane here.
- Forgetting the moment equations. Resolving alone gives four equations in six unknowns; the two moment equations are what fix the frictional forces.
- Leaving F out of the normal reaction R_1. The lower cylinder's friction pushes the upper one away from the plane, so R_1 is less than $W_1\cos\alpha$.

Beyond STEP

A rigid body is in equilibrium only if both the resultant force and the resultant moment vanish, and choosing a clever point to take moments about is the main skill of rigid-body statics. Here we had exactly as many equations as unknowns; when there are more unknowns than equations, the problem is *statically indeterminate*, and you need to model how the bodies deform to finish. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and torque carefully.

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